

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

*I G. Santosh Kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *G. Santosh Kumar*

Experiments

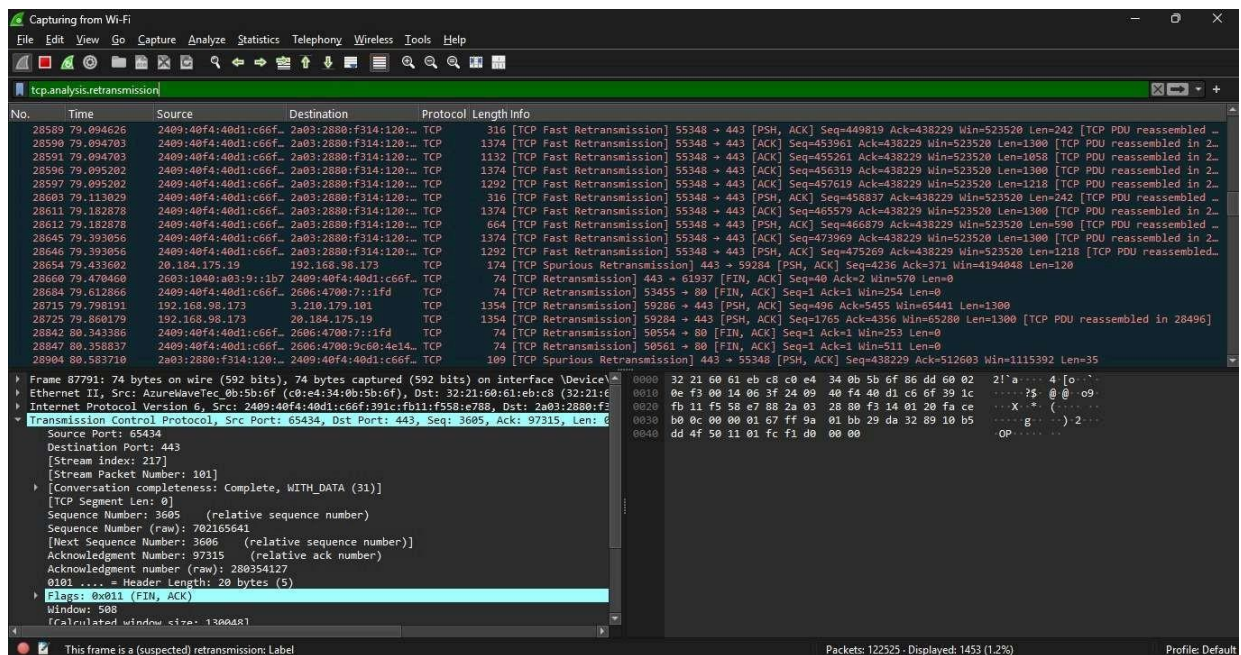
QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
87942	211.951622	2409:40f4:40d1:c66f...	2409:40f4:40d1:c66f...	DNS	178	Standard query response 0xb163 A p2p-sgpl.discovery.steamserver.net A 103.10.124.123 A 103.10.124.124 A 103.10.124.125
88001	211.958427	192.168.98.7	192.168.98.173	DNS	158	Standard query response 0xb410 HTTPS peoplestackwebexperiments-pa.clients6.google.com SOA ns1.google.com
88002	211.958427	192.168.98.7	192.168.98.173	DNS	136	Standard query response 0xb903 AAAA peoplestackwebexperiments-pa.clients6.google.com AAAA 2404:6800:4002:829::2...
88003	211.958427	192.168.98.7	192.168.98.173	DNS	124	Standard query response 0xb2d5f A peoplestackwebexperiments-pa.clients6.google.com A 192.178.177.95
88004	211.959146	192.168.98.7	192.168.98.173	DNS	144	Standard query response 0xf65b HTTPS peoplestack-pa.clients6.google.com SOA ns1.google.com
88005	211.959146	192.168.98.7	192.168.98.173	DNS	122	Standard query response 0x266e AAAA peoplestack-pa.clients6.google.com AAAA 2404:6800:4002:827::200a
88006	211.960580	192.168.98.7	192.168.98.173	DNS	118	Standard query response 0xa26d A peoplestack-pa.clients6.google.com A 142.250.182.186
89300	212.549044	192.168.98.173	192.168.98.7	DNS	83	Standard query 0x3af1 A f.c2r.ts.cdn.office.net
89301	212.549070	192.168.98.173	192.168.98.7	DNS	83	Standard query 0x8f79 AAAA f.c2r.ts.cdn.office.net
89302	212.552333	192.168.98.7	192.168.98.173	DNS	205	Standard query response 0x3af1 A f.c2r.ts.cdn.office.net CNAME office-f-net.trafficmanager.net CNAME office.mic...
89303	212.553906	192.168.98.7	192.168.98.173	DNS	229	Standard query response 0x8f79 AAAA f.c2r.ts.cdn.office.net CNAME office-f-net.trafficmanager.net CNAME office.mic...
89649	213.968098	192.168.98.173	192.168.98.7	DNS	91	Standard query 0x794c HTTPS espresso-pa.clients6.google.com
89650	213.968975	192.168.98.173	192.168.98.7	DNS	91	Standard query 0x78c5 AAAA espresso-pa.clients6.google.com
89653	213.969942	192.168.98.173	192.168.98.7	DNS	91	Standard query 0xf625 A espresso-pa.clients6.google.com
89660	214.028221	192.168.98.7	192.168.98.173	DNS	141	Standard query response 0x794c HTTPS espresso-pa.clients6.google.com SOA ns1.google.com
89672	214.028812	192.168.98.7	192.168.98.173	DNS	119	Standard query response 0x78c5 AAAA espresso-pa.clients6.google.com AAAA 2404:6800:4002:829::2...
89673	214.028812	192.168.98.7	192.168.98.173	DNS	107	Standard query response 0xf625 A espresso-pa.clients6.google.com A 192.178.158.95
89750	214.231366	192.168.98.173	192.168.98.7	DNS	98	Standard query 0xebb3 HTTPS appsgenaiserver-pa.clients6.google.com

Frame 88001: 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface \Device\NPF...
Ethernet II, Src: 32:21:60:61:eb:c8 (32:21:60:61:eb:c8), Dst: AzureWaveTc_8b:5b:6f (c8:e4:34:8b:5b:6f)
Internet Protocol Version 4, Src: 192.168.98.7, Dst: 192.168.98.173
User Datagram Protocol, Src Port: 53, Dst Port: 51563
Domain Name System (response)

Domain Name System: Protocol

Packets: 116243 - Displayed: 1100 (0.9%)

Profile: Default

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp

No.	Time	Source	Destination	Protocol	Length	Info
5484	18.166858	2600:140f:5e00:4::1...	2409:40f4:40d1:c66f...	TCP	2674	443 → 51259 [PSH, ACK] Seq=3361904 Ack=2442 Win=501 Len=2600 [TCP PDU reassembled in 5502]
5485	18.167089	2409:40f4:40d1:c66f...	2600:140f:5e00:4::1...	TCP	74	51259 → 443 [ACK] Seq=2608 Ack=3364104 Win=2047 Len=0
5486	18.171820	2600:140f:5e00:4::1...	2409:40f4:40d1:c66f...	TCP	2674	443 → 51259 [PSH, ACK] Seq=3364104 Ack=2442 Win=501 Len=2600 [TCP PDU reassembled in 5502]
5487	18.171899	2409:40f4:40d1:c66f...	2600:140f:5e00:4::1...	TCP	74	51259 → 443 [ACK] Seq=2608 Ack=3366704 Win=2047 Len=0
5488	18.192277	2600:140f:5e00:4::1...	2409:40f4:40d1:c66f...	TCP	2674	443 → 51259 [PSH, ACK] Seq=3366704 Ack=2442 Win=501 Len=2600 [TCP PDU reassembled in 5502]
5489	18.192277	64:ff9b::143:ceda1	2409:40f4:40d1:c66f...	TCP	76	443 → 51269 [ACK] Seq=1 Ack=1609328 Win=16385 Len=0
5490	18.192277	2600:140f:5e00:4::1...	2409:40f4:40d1:c66f...	TCP	2674	443 → 51259 [PSH, ACK] Seq=3369304 Ack=2442 Win=501 Len=2600 [TCP PDU reassembled in 5502]
5491	18.192359	2409:40f4:40d1:c66f...	2600:140f:5e00:4::1...	TCP	74	51259 → 443 [ACK] Seq=2608 Ack=3371904 Win=2047 Len=0
5492	18.192438	2409:40f4:40d1:c66f...	64:ff9b::143:ceda1	TCP	1374	51269 → 443 [ACK] Seq=1705362 Ack=1 Win=255 Len=1300 [TCP PDU reassembled in 5493]
5493	18.192438	2409:40f4:40d1:c66f...	64:ff9b::143:ceda1	SSLv2	1374	Encrypted Data
5494	18.194713	2600:140f:5e00:4::1...	2409:40f4:40d1:c66f...	TCP	2674	443 → 51259 [PSH, ACK] Seq=3371904 Ack=2442 Win=501 Len=2600 [TCP PDU reassembled in 5502]
5495	18.194760	2409:40f4:40d1:c66f...	2600:140f:5e00:4::1...	TCP	74	51259 → 443 [ACK] Seq=2608 Ack=3374504 Win=2047 Len=0
5496	18.209063	64:ff9b::143:ceda1	2409:40f4:40d1:c66f...	TCP	76	443 → 51269 [ACK] Seq=1 Ack=1611928 Win=16385 Len=0
5497	18.209063	64:ff9b::143:ceda1	2409:40f4:40d1:c66f...	TCP	76	443 → 51269 [ACK] Seq=1 Ack=1614528 Win=16385 Len=0
5498	18.209116	2409:40f4:40d1:c66f...	64:ff9b::143:ceda1	TCP	1374	51269 → 443 [ACK] Seq=1707962 Ack=1 Win=255 Len=1300 [TCP PDU reassembled in 5520]
5499	18.209116	2409:40f4:40d1:c66f...	64:ff9b::143:ceda1	TCP	1374	51269 → 443 [ACK] Seq=1709262 Ack=1 Win=255 Len=1300 [TCP PDU reassembled in 5520]
5500	18.209116	2409:40f4:40d1:c66f...	64:ff9b::143:ceda1	TCP	1374	51269 → 443 [ACK] Seq=1710562 Ack=1 Win=255 Len=1300 [TCP PDU reassembled in 5520]
5501	18.209116	2409:40f4:40d1:c66f...	64:ff9b::143:ceda1	TCP	1374	51269 → 443 [ACK] Seq=1711862 Ack=1 Win=255 Len=1300 [TCP PDU reassembled in 5520]

Sequence Number: 3371904 (relative sequence number)
Sequence Number (raw): 1824252171
[Next Sequence Number: 3374504 (relative sequence number)]
Acknowledgment Number: 2442 (relative ack number)
Acknowledgment Number (raw): 1912666515
0101 ... = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 501
[calculated window size: 501]
[window size scaling factor: -1 (unknown)]
Checksum: 0xa3ff [unverified]
Checksum Status: Unverified
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
TCP payload (2600 bytes)
[Reassembled PDU in frame: 5502]
TCP segment data (2600 bytes)

On-Screen Keyboard